

PROJECT	BASE DEVELOPMENT FEE	AI UPSELL RANGE	TARGET ARCHITECTURE
SeekFactory (Web + App + Backend)	₹1,50,000 INR (1.5 Lakhs)	₹45,000 – ₹1,20,000 INR	Spring Boot 4 + Qwen / DeepSeek

1. Executive Summary & Strategic Positioning

The client has requested an **Artificial Intelligence (AI) integration** for SeekFactory. In cross-border B2B manufacturing between Indian buyers and Chinese factories, standard search and basic text chat suffer from heavy friction: language barriers, specialized engineering terminology (tolerances, surface roughness, Incoterms, material grades), and buyer difficulty in formulating complete Requests for Quotation (RFQs).

Integrating domain-specific AI transforms SeekFactory from a passive catalog into an **intelligent, self-service industrial sourcing broker**. Crucially, because mainland China operates behind the **Great Firewall (GFW)**, mainstream Western APIs (OpenAI, Claude, Gemini) fail when called from Chinese IP addresses. This document outlines the exact features, legal & cross-border architecture, cost breakdown, and quotation packages to present to the client.

2. Top 5 High-Impact AI Features to Build

Feature 1: "Ask SeekFactory" — Natural Language AI Sourcing Concierge Problem: Buyers struggle navigating deep 2-tier taxonomies (187+ machinery subcategories) to find factories matching exacting industrial standards. Solution: A persistent conversational AI assistant available on web and mobile. A buyer can type or speak in plain English or Hindi: <i>"I need a factory in Zhejiang or Guangdong with 5-axis CNC milling for aerospace aluminum 7075, tolerance ±0.005mm, batch of 500 units with CMM inspection certs."</i> How it works: Leverages RAG (Retrieval-Augmented Generation) on PostgreSQL with the pgvector extension. The query is embedded, vector-searched against verified factory equipment lists and product specs, and the AI outputs matched suppliers with a 1-click <i>"Create Pre-filled RFQ"</i> CTA.	High Value · Core Sourcing
Feature 2: Real-time Technical Translation in B2B Chat (EN/HI ⇄ ZH) Problem: Indian buyers communicate in English or Hindi; factory engineers in China communicate in Mandarin Chinese. Generic translators (Google/DeepL) frequently butcher specialized mechanical terms (e.g., mistranslating "EDM wire cutting", "anodizing thickness in microns", or "draft angles"). Solution: Integrated directly into the existing InteractiveChatView. Indian buyers type naturally; the Chinese supplier instantly sees technically accurate Simplified Chinese. When the supplier replies in Mandarin, the buyer sees precise English.	Essential · High Adoption

Feature 3: AI Smart RFQ Drafter & Technical Spec Extractor

Conversion Booster

Problem: Over 60% of B2B buyers abandon RFQ forms because they do not know what Incoterms (FOB/CIF), HS Codes, or material specifications to enter.

Solution: The buyer uploads a 2D engineering drawing (PDF/image) or types rough notes. The AI extracts and auto-fills: Product Name, Suggested Category, MOQ, Recommended Incoterm, Target Currency, and builds the structured specs JSONB dictionary automatically.

Feature 4: Reels Video Auto-Transcription & Bilingual Subtitles

Video Feed Engagement

Problem: SeekFactory is video-first. Chinese plant managers narrate machine tours in Mandarin, making the video reels incomprehensible to Indian and Western buyers.

Solution: On video reel upload, the AI transcribes Chinese factory audio, translates it into synchronized English captions, and auto-generates SEO titles, technical descriptions, and hashtags (e.g., #PrecisionForging, #LaserCutting).

Feature 5: Supplier Quotation Comparison & Decision Assistant

Enterprise Decision Support

Problem: When an RFQ receives 3–5 quotes with disparate terms (different currencies, FOB vs. CIF, varying tooling charges), buyers struggle to assess the real total cost of ownership.

Solution: AI analyzes the `rfq_quotes` table and produces a comparison matrix highlighting best landed cost, fastest lead time, and supplier reliability ratings.

3. The Cross-Border Constraint: China ⇄ India Model Reality

This is the **most critical technical finding** of our project analysis. Most agencies make the mistake of using OpenAI or Google Gemini, which causes immediate failure for Chinese users.

MODEL / PROVIDER	MAINLAND CHINA SAFE?	INDIA SAFE?	TECHNICAL EN-ZH QUALITY	TOKEN COST (PER 1M TOKENS)	VERDICT FOR SEEKFACTORY
Alibaba Qwen (DashScope API)	100% Legal & Safe	Yes	★★★★★ Exceptional (Built for B2B)	~\$0.20 (₹16 / 1M tokens)	Top Choice #1 (Recommended)
DeepSeek (V3 / R1 API)	100% Legal & Safe	Yes	★★★★★ Superb Reasoning & Code	~\$0.14 (₹11 / 1M tokens)	Top Choice #2 (Best Price/Perf)
OpenAI (GPT-4o / Mini)	Blocked by GFW	Yes	★★★★ Very Good	\$2.50 – \$10.00	Avoid (Fails in China)
Self-Hosted (Ollama / vLLM)	Depends on Server	Depends on Server	★★★ Hardware Dependent	\$400–\$800/mo server cost	Avoid (Prohibitive Server Cost)

Recommended Architecture: Spring Boot Backend Proxy

Rule: The frontend (Next.js & Expo Mobile) **never** calls AI APIs directly.

- Frontend calls `/api/v1/ai/chat` or `/api/v1/ai/translate` on the Spring Boot backend.
- Spring Boot backend (hosted in a neutral hub like AWS Singapore `ap-southeast-1` or Mumbai with direct China peering) connects securely to **Alibaba Cloud Qwen** or **DeepSeek API**.
- Responses are streamed back to the client using **Server-Sent Events (SSE)** for real-time typewriter rendering.
- Result:** Zero VPN requirements for Chinese factory managers, zero blocked domains, and complete operational compliance in both countries.

4. Infrastructure Decision: Cloud API vs. Self-Hosted Ollama

EVALUATION METRIC	CLOUD API (DEEPSEEK / ALIBABA QWEN)	SELF-HOSTED (OLLAMA ON DEDICATED GPU)
Monthly Hosting Cost	₹500 – ₹2,000 / month (Pay only for tokens used)	₹35,000 – ₹65,000 / month (AWS EC2 g5.xlarge / A10G GPU)
Implementation Effort	Fast: 2–3 days using Spring Boot HTTP WebClient	Slow: 2–3 weeks for CUDA, VRAM tuning, model quantization
Maintenance & Reliability	Zero maintenance, 99.9% uptime managed by provider	High maintenance: Out-of-Memory crashes, concurrency bottlenecks
Recommendation	100% Recommended for this project budget	Reject — server costs will destroy client margin

5. How Much to Charge: Client Quotation Packages

Because you charged **₹1,50,000 (1.5 Lakhs)** for the base platform (Next.js + Expo Mobile + Spring Boot), AI represents an enterprise capability upgrade. You should present three structured tiers:



Starter AI ₹45,000 – ₹60,000 Search Concierge & B2B Chat Translation ✓ "Ask SeekFactory" RAG AI Assistant ✓ Database Vector Search on Products & Plants ✓ Real-Time Technical Chat Translation (EN ↔ ZH) ✓ Spring Boot AI Service & SSE Streaming ✓ Timeline: ~1.5 to 2 Weeks Base upgrade package	Pro Sourcing AI RECOMMENDED ₹85,000 – ₹1,10,000 Full Sourcing Suite + Smart RFQ ✓ Everything in Starter AI ✓ AI Smart RFQ Auto-Drafter from Specs ✓ Drawing & Specs JSONB Extractor ✓ Video Reels Auto-Tagging & Descriptions ✓ Dual-Language Industrial Prompt Chips ✓ Timeline: ~3 to 4 Weeks Best margin & value balance	Enterprise AI Suite ₹1,40,000 – ₹1,80,000 Autonomous Sourcing & Video Subtitles ✓ Everything in Pro Sourcing AI ✓ Supplier Quote Comparison Engine ✓ Video Reels Audio Subtitling (Mandarin to EN) ✓ Supplier Capability Verification Scoring ✓ Priority Prompt Engineering & SLA ✓ Timeline: ~5 to 6 Weeks Full platform transformation
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💡 **Negotiation Strategy with the Client:**

Pitch **Package 2 (Pro Sourcing AI) at ₹95,000 INR**. If the client attempts to negotiate or hesitates on budget, gracefully close on **Package 1 (Starter AI) at ₹50,000 INR**. This yields an instant **33% to 63% increase** in overall project revenue with manageable technical overhead.

6. Technical Implementation Roadmap & Risk Mitigation

- **PostgreSQL pgvector Integration:** Enable `CREATE EXTENSION vector;` in PostgreSQL 15+. Store 1536-dimensional embeddings directly in a `manufacturer_embeddings` table. Avoids recurring SaaS costs of Pinecone or Weaviate.
- **Hallucination Prevention Guardrails:** Industrial buyers cannot tolerate false claims. The system prompt will strictly mandate: *"Only recommend manufacturers present in the verified context database. If no manufacturer matches the requested tolerance or material, state this clearly and provide the custom RFQ link."*
- **Streaming Response UX:** Implement Spring WebFlux / SSE endpoints so responses render with real-time typewriter animations on both Next.js and Expo React Native.